Telco Customer Churn Analysis Report

1.Title

Telco Customer Churn Analysis
Customer Retention & Churn Risk Analysis using Excel

Objective:

To identify the key drivers of customer churn, identify high-risk customer segments, and provide data-driven recommendations to improve customer retention.

2. Problem Statement

Customer churn occurs when a customer stops using a company's services.

For a telecom company, customer churn is important because losing existing customers can reduce recurring revenue and increase the need to acquire new customers.

The objective of this analysis is to:

- Understand the overall churn level.
- Compare churned and retained customers.
- Identify the major factors associated with churn.
- Identify high-risk and loyal customer segments.
- Analyze contract type, monthly charges and customer tenure.
- Develop actionable retention strategies.

3. Dataset Description

The **Telco Customer Churn dataset** contains customer-level information related to demographics, services, contracts, payment methods, tenure and churn status.

Important variables analyzed

Category	Variables
Customer Information	Customer ID, Gender, Senior Citizen
Tenure	Tenure / Customer lifetime
Contract	Month-to-month, One year, Two year
Charges	Monthly Charges, Total Charges
Services	Internet Service, Phone Service
Payment	Payment Method
Target Variable	Churn

The analysis focuses primarily on **Churn**, **Contract**, **Tenure**, **Monthly Charges**, **Internet Service**, **Payment Method**, and **Senior Citizen status**.

4. Approach / Methodology

The analysis was performed through the following steps:

1. Data Cleaning

- Removed/handled missing values.
- Checked data types.
- Standardized categorical fields.
- Prepared data for analysis.

2. Exploratory Data Analysis

- Compared churned vs retained customers.
- Calculated churn rate.
- Analyzed contract-wise churn.
- Analyzed tenure-wise churn.
- Examined monthly charges.
- Examined customer segments.

3. Segmentation
Customers were divided into:

- High-risk customers
- Medium-risk customers
- Loyal/low-risk customers

4. Dashboard Development
Power BI was used to create:

- KPI cards
- Churn-rate analysis
- Contract analysis
- Tenure analysis
- Customer segmentation
- Interactive visualizations

5. Analysis / Model

Key KPIs

KPI	Result
Total Customers	**7,032**
Churned Customers	**1,869**
Retained Customers	**5,163**
Overall Churn Rate	**26.57%**
ARPU	**₹64.79**

Churn Rate

The overall churn rate is **26.57%**, meaning approximately **1 in every 4 customers** has churned.
This indicates that customer retention is an important business issue.

6. Key Drivers of Churn

A. Contract Type

Contract	Churn Rate
Month-to-month	**42.71%**
One year	**11.28%**
Two year	**2.85%**

Insight

Contract type is one of the strongest indicators associated with churn.
Month-to-month customers have a **42.71% churn rate**, compared with only **2.85%** for two-year customers.
This suggests that customers without long-term commitments are considerably more vulnerable to churn.

B. Customer Tenure

The highest concentration of churn occurs among customers with **0–10 months of tenure**.

Tenure	Churned Customers
0–10 months	**968**
11–20 months	283
21–30 months	174
31–40 months	141
41–50 months	115
51–60 months	95
Above 60 months	93

Insight

The 0–10 month segment contributes approximately **51.8% of all churned customers**.
This indicates that the **early customer lifecycle is a critical retention period**.

C. Internet Service

Among churned customers:

Internet Service	Share of Churned Customers
Fiber Optic	69.40%
DSL	24.56%
No Internet	6.05%

Insight

Fiber-optic customers represent the largest share of churned customers.

However, this should be interpreted carefully: **a high share of churn does not automatically mean fiber optic causes churn**. The underlying customer population and churn rate within each service category should also be considered.

D. Payment Method

Payment Method	Share of Churned Customers
Electronic Check	**57.30%**
Mailed Check	16.48%
Bank Transfer	13.80%
Credit Card	12.41%

Insight

Electronic-check users represent the largest share of churned customers.

This segment should be investigated further to determine whether payment experience, billing issues, or customer characteristics are contributing to churn.

E. Senior Citizen Segment

Segment	Churn Rate
Senior Citizen	**41.68%**
Non-Senior Citizen	**23.65%**

Insight

The senior-citizen segment has a substantially higher churn rate than the non-senior segment.

This indicates an opportunity for more personalized customer support and retention programs.

7. Customer Segmentation

High-Risk Customers
A high-risk customer profile can be defined as customers with combinations such as:

- Month-to-month contract
- Low tenure
- Higher monthly charges
- Fiber-optic service
- Electronic-check payment
- Senior-citizen segment

A customer possessing several of these characteristics should receive greater retention attention.

Medium-Risk Customers

Examples:

- Moderate tenure
- Month-to-month or one-year contract
- Moderate/high monthly charges
- Some additional services

These customers should receive regular engagement and targeted offers.

Loyal Customers

Typical characteristics:

- Long tenure
- Two-year contract
- Stable payment method
- Lower churn-risk profile
- Longer customer relationship

These customers should be targeted with **loyalty programs and retention benefits**.

8. Visualizations

Your Power BI report should ideally contain these visuals:

1. **KPI Cards**
    - Total Customers
    - Churned Customers
    - Churn Rate
    - ARPU
2. **Churn vs Retained**
    - Donut/Pie chart
3. **Contract vs Churn Rate**
    - Column chart
4. **Tenure vs Churn**
    - Column/line chart
5. **Monthly Charges vs Churn**
    - Box plot or column chart
6. **Internet Service vs Churn**
    - Bar chart
7. **Payment Method vs Churn**
    - Bar chart
8. **Senior Citizen vs Churn**
    - Column chart
9. **Churn by Tenure Bucket**
    - 0–10
    - 11–20
    - 21–30
    - 31–40
    - 41–50
    - 51–60
    - Above 60
10. **Interactive Slicers**

- Contract
- Internet Service
- Payment Method
- Senior Citizen
- Tenure

9. Key Insights

Insight 1 — Contract is a major churn indicator

Month-to-month customers have a **42.71% churn rate**, while two-year customers have only **2.85%**.

Insight 2 — Early-tenure customers are most vulnerable

Customers with **0–10 months of tenure account for approximately 51.8% of churned customers**.

Insight 3 — Senior customers show higher churn

Senior citizens have a **41.68% churn rate**, considerably higher than the **23.65%** rate among non-senior customers.

Insight 4 — Fiber-optic customers dominate churn volume

Fiber-optic customers account for **69.40% of churned customers**, making this segment important for further investigation.

Insight 5 — Electronic-check customers require attention

Electronic-check users account for **57.30% of churned customers**, suggesting that this customer group should be examined for billing/payment-related issues.

Insight 6 — Long-term contracts are associated with retention

Customers with longer contracts show substantially lower churn than month-to-month customers.

10. Retention Strategy Recommendations

1. Target customers during the first 10 months

Create an **early-life retention program**:

- Welcome campaigns
- 30/60/90-day follow-ups
- Customer satisfaction checks
- Personalized offers
- Early issue resolution
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2. Encourage long-term contracts

Offer appropriate incentives for customers moving from:

Month-to-month → One-year → Two-year

This could improve customer commitment and reduce churn.

3. Investigate fiber-optic churn

Analyze:

- Pricing
- Service quality
- Complaints
- Network issues
- Customer expectations
- Competitor pricing

before deciding on corrective action.

4. Improve payment experience

For electronic-check customers:

- Simplify payment processes.
- Provide billing reminders.
- Investigate failed/late payments.
- Encourage alternative automatic payment options where appropriate.

5. Create targeted senior-customer programs

Provide:

- Dedicated support
- Simple communication
- Personalized assistance
- Loyalty benefits

6. Build a churn-risk dashboard

The next stage would be to create a **Customer Churn Risk Score** so that the business can identify customers who are likely to leave **before they actually churn**.

11. Conclusion

The Telco Customer Churn analysis shows that customer churn is not evenly distributed across the customer base.

The strongest patterns are associated with **contract type, customer tenure and specific customer/service segments**. Month-to-month customers and newer customers represent particularly important retention opportunities.

The analysis suggests that the company should move from a **reactive retention strategy**—trying to win customers back after they leave—to a **proactive strategy** that identifies high-risk customers early.

By focusing on early-tenure customers, encouraging long-term contracts, investigating high-churn service segments and providing targeted retention campaigns, the company can potentially improve customer retention and protect recurring revenue.

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